Arunabha Majumder

+45 71 62 24 00 | Aalborg, Denmark
somrkmv1997@gmail.com | 97arunabhasit027@gmail.com
Portfolio: arunabha22.github.io/arunabha-majumder

SE3 Labs, Hardware Development, Munich

Mechanical Design Engineer, Unmanned Systems

Dear SE3 Labs team,

I'd like to apply for the Mechanical Design Engineer, Unmanned Systems role. Designing mechanisms and electromechanical systems, building them fast and iterating until they work is exactly how I like to work, and owning the physical integration of a whole system would be a great next step for me.

As a research assistant at Aalborg University I designed a hybrid shoulder exoskeleton. I started from the torque needed to lift the arm and designed a variable stiffness mechanism, with a worm gear that pretensions the spring, working in parallel with a 6 Nm motor. I optimised the mechanism with a genetic algorithm (diameter, spring stiffness and pretension force), designed the parts, interfaces and drawings in SolidWorks, 3D printed prototypes, and had the final parts manufactured. I integrated the motor, encoder and force/torque sensors and verified its stiffness on a physical test setup. It uses over 25% less motor energy and won the Best Paper Award at ISRM 2026.

I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM), estimated its stiffness with less than 10% error and quantified human arm impedance, so kinematics and force calculations are part of my daily work. In my Master's I worked in a team of 4 electronics and mechanical engineers on a soft exosuit, and in my Bachelor's I did the CAD, machining and assembly of an arm exoskeleton and exo glove, and built a Go-Kart and a hybrid Efficycle with student teams. I also hold a Senior Diploma in Fine Arts and do graphic design (Illustrator, Photoshop, Procreate), which helps me sketch concepts and explain trade-offs clearly.

I'm ready to move to Munich and I'm willing to learn German. I would love to bring new and innovative ideas to your unmanned systems and grow with your hardware team. My portfolio, with videos and results from my work, is at arunabha22.github.io/arunabha-majumder. Thank you for reading.

Best,
Arunabha Majumder